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Cognitive and metacognitive support in learning with a serious game about demographic change

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ABSTRACT

The goal of this experiment was to investigate the impact of metacognitive training, and metacognitive and cognitive prompting, in a game-based simulation related to demographic change. The work here is based on supporting learning with serious games by scaffolding students' self-regulated learning by fostering metacognition in constructivist learning environments. The impact of metacognitive training, and/or cognitive and metacognitive prompting, on learning outcomes and cognitive load was analyzed. A pre- and post-test assessing knowledge acquisition, metacognitive behavior, and cognitive load were applied. The learning environment was a game-based simulation of a small village that was confronted with different problems related to demographic change. Learners were tasked with developing sustainable solutions to keep the village prospering. With regard to knowledge acquisition, metacognitive and cognitive prompting proved effective, while direct metacognitive training only had a minor effect. The expected interaction effect did not turn out to be statistically significant, nor did the results show a reduction of cognitive load through direct or indirect metacognitive and cognitive support. Taken together, outcomes suggest here that game-based learning with simulations can be supported by direct and indirect metacognitive and cognitive support.

1. Introduction

Game-based learning with simulations and/or serious games has become an increasingly popular approach in today's science classroom (e.g., Djaouti, Alvarez, & Jessel, 2011). Its objective is to motivate learners and increase learning performance (e.g., Egenfeldt-Nielsen, 2007). One of the core features of these digital learning opportunities is interactivity. It is assumed to foster active learning that demands different self-regulated learning processes in contrast to learning from rather passive media. Learning with simulations or games requires interaction with the learning environment in order to generate, select, organize, and integrate information, while in learning environments with a rather passive information access, information is presented in a given sequence and cannot be changed (cf. (Mayer, 2014; Wirth & Leutner, 2006)). When referring to game-based learning and simulations, it must be pointed out that there is no clear-cut distinction between the two categories. A core element of both approaches is that they are interactive and digital (Vogel et al., 2006; Wouters & Van Oostendorp, 2013). Vogel et al. (2006, p. 231; cf. Clark, Tanner-Smith, & Killingsworth, 2014) nevertheless provide a definition that makes a distinction between games and simulations: "A computer game is defined as such by the author, or inferred by the reader because the activity has goals, is interactive, and is rewarding (gives feedback). Interactive simulation activities must interact with the user by offering the options to choose or define parameters of the simulation then observe the newly created sequence rather than simply selecting a prerecorded simulation." However, these definitions present two rather extreme perspectives, because elements of simulations such as parameter definitions and the observation of the impact of parameter changes on dependent variables might also be implemented in computer games. In addition, there is a wide range of applications regarded as "simulations" or components thereof, which may also include prerecorded sequences. Sitzmann (2011, p. 490) suggests to combine simulations and games under the term "simulation games", and provides the following definition: "Recently, the term 'serious games' was coined to refer to simulation games designed to address more complicated and thought-provoking issues". A core element here is that learners have to make decisions in a digital learning environment and learn about the consequences of the decisions they make during gameplay (Clark et al., 2014; Sitzmann, 2011).

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The theoretical bases for most research on games and simulation-based learning are constructivism and humanism (Wu, Hsiao, Wu, Lin, & Huang, 2012). This research mostly refers to principles of experiential learning, situated learning theory, and problem-based learning. Key reasons for using games for learning are summarized by Myers and Reigeluth (2017): games can provide authentic practice, and thus provide the link between action and cognition; they also provide a huge potential for team development, social learning, and social cohesion, can enhance learner engagement and effort, provide a safe environment for learning, and are adaptable to learner characteristics (Vogel et al., 2006). Simulation-based games enable and encourage learners to explore and interact with a digital learning environment by changing parameters and observing the results of this interaction (Esquembre, 2002). Nevertheless, game-based learning environments do not necessarily contribute to enhanced knowledge acquisition (Wouters & Van Oostendorp, 2013). Kim, Park, and Baek (2009) suggest six relevant processes learners need to follow: (1) learners have to identify one or more problems and its/their relevant dimension(s) to be solved during gameplay. (2) In addition, learners must generate different alternatives for action, (3) choose a problem-solving strategy, (4) react to feedback from the learning environment and use it, (5) keep on using successful strategies, and (6) change strategies when they prove unsuccessful.

Supporting learners within game-based learning environments is a core instructional approach to enhance knowledge acquisition and problem-solving. The meta-analyses provided by Wouters and Van Oostendorp (2013) revealed that additional instructional support significantly increases learning in game-based learning environments. This effect mainly relates to support strategies for learners in selecting new information. Especially metacognitive strategies appear to be essential for successful learning in game-based learning environments. In their study, Kim et al. (2009) examined three metacognitive strategies and their impact on performance in a game-based learning environment (in the domain of economy). The authors compared written self-recording, modeling, and thinking-aloud strategies. The results revealed that especially modeling and thinking aloud were effective strategies for fostering learning and keeping students involved in the game. Another study provided by Riemer and Schrader (2016) found that learning with and from a serious game is enhanced, when learners show a high degree of self-monitoring during gameplay.

Sung and Hwang (2013) investigated the impact of collaborative vs. individual game-based learning, and the impact of an external representation tool for supporting and planning the learning process. Using a science-focused scenario, authors found that collaborative learning by means of the game, supported by the external representation tool, led to the highest level in learners' knowledge acquisition, compared to collaborative learning without the external representation device and individual learning with the device. All in all, the outcomes of this study reveal that planning and monitoring game-based learning supported by collaborative learning, and external scaffolding, contributes to learning success, while a combination of both approaches was the most efficient intervention in this respect. However, there is also strong evidence that collaborative game-based learning per se does not automatically lead to better learning outcomes. The study by Meluso, Zheng, Spires, and Lester (2012) as well as the meta-analysis by Clark et al. (2014) do not find beneficial effects of collaborative playing compared to individual use.

Another predictor for the success of game-based learning is to challenge learners. Hamari, Shernoff, Rowe, Coller, et al. (2016) found that the greater the perceived challenge of a game, the more learners engage, and the higher their learning outcomes. As a consequence, authors suggest that the challenge of games should be adjusted to increasing abilities and skills of learners. These findings support the assumption that "simply playing" educational games does not automatically lead to a persistent level in motivation or elaboration processes. Learners benefit more from game-based learning when additional instructional support is provided. Erhel and Jamet (2013) showed in this respect that providing a rather instructional focus leads to deeper learning compared to a rather entertaining focus as framed by the instruction. Nevertheless, providing regularly feedback to students that have a rather focus on the entertaining aspect of a digital game-based learning environment also show deeper elaboration processes as triggered by the feedback.

Taken together, these findings show that game-based learning benefits from additional instructional support (e.g., Wouters & Van Oostendorp, 2013). Nevertheless, there is evidence that this support should be part of the learning environment itself and not be separated (e.g., by providing additional lecturing; Clark et al., 2014).

The findings as described here also provide evidence that especially fostering cognitive and metacognitive processes supports knowledge and skill acquisition in game-based self-regulated learning.

1.1. *Supporting self-regulated learning in digital learning environments*

Learning with simulations or games is primarily self-regulated (Eckhardt, Urhahne, Conrad, & Harms, 2013). This self-regulated learning might be beneficial if learners are challenged and motivated (Erhel & Jamet, 2013), but it might be ineffective if students are unable to regulate their learning activities (Bannert & Mengelkamp, 2013). Here they need additional guidance or support. One possibility to support students' self-regulated learning is to foster the use of metacognitive strategies, either by direct instruction or more indirect prompting during learning. Research providing approaches to fostering metacognitive processes are rather beneficial for learning, especially within complex learning environments. In a study by Barzilai and Blau (2014), the influence of scaffolding during playing a serious game was examined. The authors found a minor positive impact of the scaffolding introducing basic concepts and variables of the game. Nevertheless, the scaffolding as used here could rather be described as some kind of pre-gameplay training rather than a truly supportive strategy for self-regulated learning during the game. There is evidence that students also need assistance during the learning process and progress. According to De Jong and van Joolingen (1998), learners often fail to apply systematic goal-oriented strategies during simulation-based learning. This is also a problem in other domains such as hypermedia learning, where learners are often faced with problems related to self-regulating their learning, mostly due to a lack of use of metacognitive activities (Bannert, 2007; Bannert, Hildebrand, & Mengelkamp, 2009; Lawless & Brown, 1997). While students might have knowledge about metacognitive learning strategies, in many cases this knowledge is rather inert and thus difficult to apply (Artelt, 1999; Bannert et al., 2009). Especially when faced with a host of possible decisions like navigation tasks in hypermedia learning environments, or possible interactions within simulations, the application of adequate metacognitive strategies might support learners' progress.

Such learning strategies include motivational strategies, cognitive, and metacognitive strategies, enabling learners to stay on a learning task and benefit from a learning environment by means of deeper elaboration strategies and meaningful as well effective information processing. This research focuses on metacognitive strategies. Being aware that students often fail to apply metacognitive strategies, it is necessary to provide guidance or clear instructions for students to use and apply metacognitive learning strategies. With regard to learning strategies in general, Bannert (2007) differentiates between three categories, namely cognitive strategies, metacognitive strategies, and resource management. Each of these categories includes a set of different strategies that can be applied during learning processes (e.g., elaboration or integration processes as cognitive strategies, cf. Wild, 2005; self-motivation, attention control, or time management as examples for resource management strategies; Kuhl, 1987). A clear and unambiguous definition for metacognition is difficult to find. According to Flavell (1981, p. 37), it is a "fuzzy concept" which is defined differently depending on the research domain (cf. Moser et al., 2017)). Weinert (1987) defines metacognition as "(…) second-order cognitions: thoughts about thoughts, knowledge

about knowledge, or reflections about actions" (p. 8). Metacognitive thoughts are deliberate, tactical, intentional, goal-directed, and future-oriented mental behaviors that can be used to accomplish cognitive tasks (Flavell, 1981). Dubs (2009) further distinguishes between declarative and executive metacognitive processes. Declarative metacognitions include knowledge about a person (e.g., one's abilities), the task (e.g., its nature), and strategies. Executive metacognitions include procedural knowledge about regulating and monitoring progress (cf. Breuer, Ettmüller, Schu, & Tauschek, 2011; Kluwe & Schiebler, 1984).

After Flavell (1981) postulated that metacognitions lead to smarter and more efficient learning, and, thus, to increased learning performance, several studies tried to confirm this relationship. While most of this research assumed that there is a positive correlation between the use of metacognitive strategies and learning success (Sperling, Richmond, Ramsay, & Klapp, 2012), empirical evidence is still heterogeneous, and provides only marginal or moderate correlations and hardly any evidence for metacognitive strategy use as a strong and valid predictor for academic performance. Schiefele, Streblow, Ermgassen, and Moschner (2003), for example, found only moderate correlations among some metacognitive strategies (organization, monitoring, self-regulation, and effort strategies) on university students' academic performance (cf. Schiefele, 2005). Similar findings are reported by Schiefele et al. (2003) who found only weak correlations among metacognitive knowledge and knowledge acquisition. They state (p. 68): "These correlations are rather low and indicate no meaningful relationship between achievement and metacognition in this sample." While this research focuses on the use of metacognitive strategies in general, there is evidence that some single strategies influence specific variables with regard to learning performance. The study by Artelt, Neuenhaus, Lingel, and Schneider (2012) found moderate correlations between reading/text comprehension and metacognitive knowledge. Veenman, Prins, and Elshout (2002) showed that metacognitive abilities can positively influence the quality (e.g., the quality of elaboration processes) of acquired knowledge, while the correlation with overall academic performance still remains weak. Artelt (1999) was able to show that especially strategic planning behavior can positively influence knowledge acquisition (cf. Bannert, 2005; Lin & Lehmann, 1999). Schiefele et al. (2003) provide evidence that the lack of metacognitive learning strategies as a predictor of academic performance can be explained by the nature of learning environments and tasks. Such strategies seem to be particularly beneficial for tasks with high degrees of freedom for accomplishing objectives. Another explanation for a lack of general effects could be derived from a lack of transfer of learning strategies from one situation to another (Garner & Alexander, 2015). They state (p. 149): "Students might conclude erroneously that the skimming and rereading strategies that they have been taught to use in long social studies text segments will not work in the single-paragraph science text that they encounter on the standardized test."

Taken together, research on metacognitive self-regulation reveals that the use of such strategies is not a matter of course. In addition, knowledge about metacognitive learning strategies does not automatically lead to their application and thus to deeper or better learning. Therefore, different approaches for fostering awareness about and use of (metacognitive) learning strategies have been developed. Such approaches include direct instruction programs or indirect mechanisms such as scripting, scaffolding, or prompting (Bannert, 2009; Friedrich & Mandl, 1997). Ideally, both training and prompting measures take place to support effective self-regulated learning, and to compensate disadvantages of each (Dubs, 2009; Friedrich & Mandl, 1997).

Training is aimed at conveying metacognitive strategies, their possible applications, and their effects (Bannert, 2004). Direct training helps learners develop effective learning strategies (Mandl & Friedrich, 1992). Research on such direct training reveals promising results in terms of the benefits of such approaches. Lucangeli, Galderisi, and Cornoldi (1995) demonstrated that training increases metacognitive knowledge and self-control over one's own learning progress (Fritz & Hussy, 2001). Yuill and Oakhill (1991) found positive effects of metacognitive training on text comprehension (cf. Carretti, Caldarola, Tencati, & Cornoldi, 2014). However, students often already possess metacognitive skills but are unable to transfer their knowledge (Bannert et al., 2009; Garner & Alexander, 2015). Here, indirect scaffolding approaches like prompting might support learners' usage of metacognitive learning strategies. According to Bannert (2004, p. 2), such prompts are "(…) instructional measures integrated in the learning context which ask students to carry out specific metacognitive activities." By providing such instructional prompts, students are not taught new information, but are rather required to activate their (prior) knowledge and apply their skills within a given context (cf. Bannert, 2009). This also draws learners' attention to crucial aspects of the learning process and makes them use their metacognitive skills, which they would otherwise not have put to spontaneous use (Artelt, 1999; Bannert, 2009; Bannert et al., 2009). Research about the effectiveness of prompting shows different benefits. Chi, Bassok, Lewis, Reimann, and Glaser (1989) found that students with prompting used self-declarations more often and achieved better learning outcomes than students without prompting. Bannert and Mengelkamp (2013) as well as Sonnenberg, Mengelkamp, Loudwin, and Bannert (2015) revealed that metacognitive prompting enables students to activate several learning strategies. Nevertheless, it has to bet noted that "metacognitive prompting" as a term might also be misleading. Prompting as such can fulfil two functions: 1) the genuine activation of metacognitive thinking processes (e.g., monitoring one's own attentional processes), and 2) acting as a metacognitive process itself, which activates subsequent cognitive processes. Thus, metacognitive prompting or scaffolding enables metacognitive and cognitive processes as well. With regard to its efficiency, some research indicates that metacognitive prompting can contribute to learning success. Chang, Chen, Lin, and Sung (2008) examined the impact of supportive scaffolds during simulation-based learning on learning success. The authors found that the integration of experiment and hypothesis prompting during simulation-based learning about optics has a greater effect on learning success than step guidance. This applies in particular to learners with stronger abstract reasoning skills. However, the authors only found small to medium effects. Zacharia et al. (2015) found that metacognitive support was useful in computer-supported inquiry learning about electric circuits compared to scaffolding content choices (cf. Borgenheimer & Weber, 2010). Kim and Pedersen (2011) found that metacognitive scaffolds (e.g., self-questions) during learning with simulations in biology led to better performance in students' hypothesis development in comparison to their performance without support. Piksööt and Sarapuu (2015) demonstrated that students' knowledge transfer can be enhanced by implementing question prompts during learning with web-based models of molecular genetics. With regard to the timing of the prompting, the study by Thillmann, Künsting, Wirth, and Leutner (2009) showed that prompting during the learning phase is more beneficial compared to prompting before the learning phase. In a study by Moser et al. (2017)opt_INS>, authors found no effect for direct metacognitive training prior to learning with a simulation in the domain of physics. However, they found a positive effect for metacognitive scaffolding during experimenting with the simulation if learners used the prompts accordingly. Although there is further evidence that metacognitive prompting supports knowledge acquisition (e.g., Lin & Lehmann, 1999; Peters & Kitsantas, 2010; Stark, Tyroller, Krause, & Mandl, 2008), some studies have shown only weak or no effects of metacognitive prompting (e.g., Bannert, 2007; Glogger, Schwonke, Holzäpfel, Nückels, & Renkl, 2008).

Overall, it remains unclear under which conditions and in which domains metacognitive prompting might be applicable to and beneficial for learning. Especially from a cognitive load perspective, the provision of an additional task during learning (e.g., metacognitive prompting) might reduce learners' capacity of working memory and, thus, inhibit rather than foster learning. From a cognitive load perspective, the use of

metacognitive and cognitive strategies during learning might generate a dual task situation that negatively influences learning. Nevertheless, there is some evidence that a) direct metacognitive training approaches, and b) metacognitive and cognitive prompting can positively contribute to learning performance when used properly, and the domain of learning provides a certain amount of complexity where the use of metacognitive and cognitive strategies supports coherence building processes and thus understanding. In addition, research findings also indicate that prompting can support executive processes of strategies acquired previously. This leads to the assumption that imparting metacognitive strategies prior to a learning phase in which the application of these strategies is supported by prompts should greatly contribute to learning performance. This might also be influenced by learners' prior knowledge regarding metacognitive abilities, leading to the (re)activation of corresponding metacognitive schemata by means of training or prompting. For this research, the domain of demographic change was chosen. The major reason here was that this topic is a substantial part in Austrian High School geography and economics education and instructional research here is rare. There is some research but mostly limited to population pyramids (e.g., Barreto, 2018; Wicks, 1974). The research here aims on integrating three perspectives: learning with serious games, supporting self-regulated learning by fostering metacognitions and fostering learning about demographic change. Taken together the interface of these three parts, research shows that some research questions remain unanswered and are subject of the present study.

The first question addresses the issue of how different strategies for fostering metacognitive and cognitive processes during game-based learning with a simulation can support knowledge building processes. It can be assumed that learners do not automatically possess adequate cognitive and metacognitive strategies for learning efficiently in such a learning environment. Teaching metacognitive strategies in advance should therefore provide them with the required skills and lead to increased learning performance (Hypothesis 1). It is also likely that learners' metacognitive skills are not automatically activated or transferred. Support in the form of prompting during learning with a simulation should lead to the application of the right learning strategies and, if used properly, to increased knowledge acquisition (Hypothesis 2). It can also be assumed that learners might not automatically activate and deploy the required skills and prior knowledge about learning strategies while learning. Thus, an interaction effect between training prior to a learning phase and the provision of prompting during learning might occur, providing the best instructional support with enhanced knowledge acquisition and reduced cognitive load (Hypothesis 3). In addition, the influence of prior domain knowledge but also prior knowledge of metacognitive and cognitive learning strategies is examined.

2. Method

The objective of this experiment was to investigate the impact of metacognitive training and metacognitive and cognitive prompting in a game-based simulation on the subject of demographic change. The impact of such metacognitive training and/or metacognitive and cognitive prompting on learning outcomes and cognitive load was analyzed.

2.1. Sample

A total of 131 high school students aged between 14 and 17 (*mean age* = 15.43; *SD* = 0.75) voluntarily participated in this experiment. 73 participants were female, 57 male (one missing value). No reward was given and each participant was randomly and individually assigned to one of the experimental conditions. The overall duration of the experiment was 90 min. The sample size was calculated *a-priori* using the G*Power software package 3.1.9.2 (Faul, Erdfelder, Lang, & Buchner, 2007). Aiming on an average effect size of 0.1, with 0.05 Alpha Error probability and a Power of 0.95, a resulting minimum sample size of 108 participants for the underlying design has been computed. The study itself was conducted in computer rooms with each student working alone at a desktop computer on the task resulting from the assigned condition.

2.2. Design

There were two independent variables in this study: 1) provision (or omission) of direct support in the form of training on metacognitive strategies prior to learning with a game-based digital learning environment, and 2) provision (or omission) of indirect support in the form of metacognitive prompting during learning with the simulation. The dependent variables were knowledge acquisition and self-reported cognitive load after working with the software. As control variables, prior knowledge, and use of cognitive and metacognitive learning strategies were assessed.

3. Material

3.1. Learning environment

The learning environment "Save the village of Bauxdorf" (original German language title: "Rettet Bauxdorf") was specially designed for this experiment. It is a game-based learning simulation that represents a highly complex dynamic system with a high degree of authenticity (cf. Wirth & Leutner, 2006). The simulation was designed to cover with the topic of demographic change. Demographic change is a topic that is dealt with in different curricula and education standards across all German-speaking countries, typically (but not exclusively) in geography lessons. Interestingly, there is hardly any work on pedagogical content knowledge and/or empirical research within this area. What is more, non-commercial educational resources are rare and mostly cover a broader area within demography (e.g. Alustiza Galarza & Lines, 2018). Demographic change and demography include a range of topics such as the population pyramid, different kinds of demographic indicators, migration flows, and their implications on society, economy, environment, work, etc. (Max-Planck Society, 2018). A major objective here was to present an authentic scenario enabling learners to discover different dynamic processes in relation to demographic change. Finding solutions to the different problems presented here requires learners to define each problem and understand the complexity and different solution approaches that might be required as a result of conflicting interests of different protagonists. Learners were tasked with developing sustainable solutions to keep the village of Bauxdorf prospering. Bauxdorf, a former mining village, is presented as a rural settlement affected by a dramatic decrease in inhabitants, whose mayor is trying to ensure the prosperity of the community. The game is set within a community council where different approaches to rescuing the village can be initiated and the consequences can be observed. All background information required in order to develop and apply adequate solutions is incorporated in the game. This includes general information about demographic change and general statistical data specific to the village of Bauxdorf (see Fig. 1).

After retrieving the relevant information, learners had to make decisions on the future of Bauxdorf. They were asked to select one out of several options, each with a different degree of effectiveness. Participants were given detailed system-generated feedback on their decisions, as well as the opportunity to change or modify their problem-solving strategy and receive renewed system-generated feedback demonstrating the impact of these decisions on the development of Bauxdorf (see Fig. 2).

The game was directly developed for this study and was based on the objectives provided by the curriculum of the Austrian Federal Ministry of Education for the 8th grade of High School for Geography and Economy. All information necessary for students to make decisions within the game were directly provided within the game environment itself. The learning environment was based on HTML and did not need

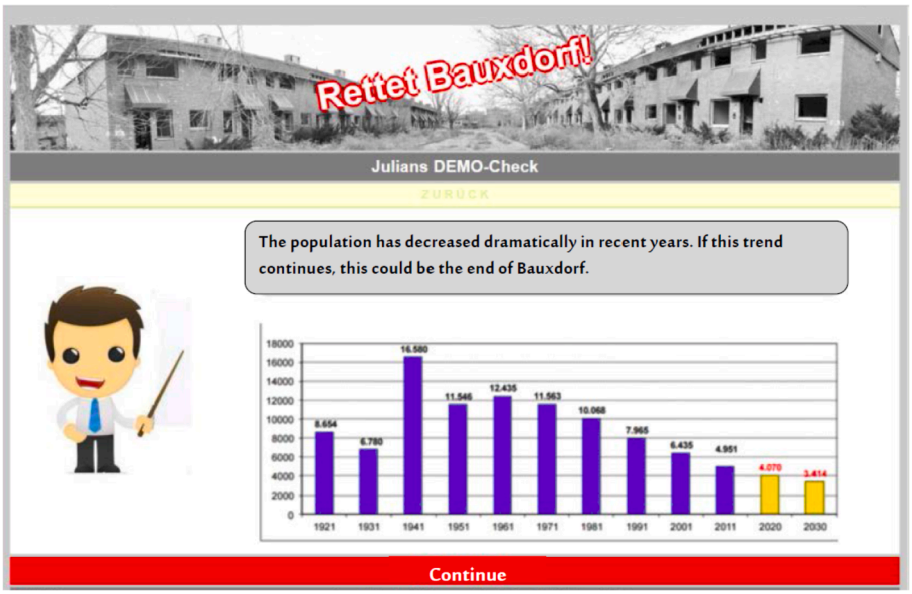


Fig. 1. General information about demographic change and general statistical data for the village of Bauxdorf.

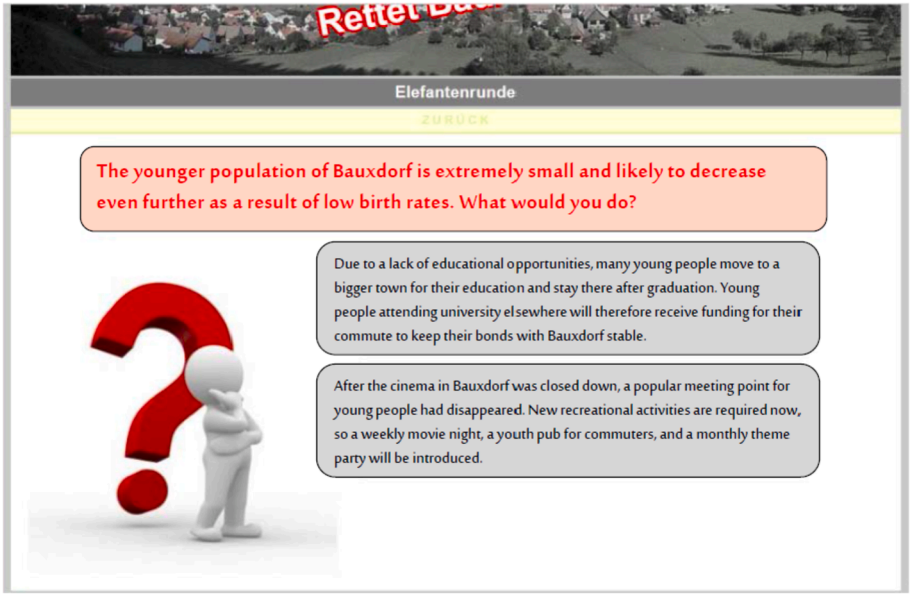


Fig. 2. Learners' choice of actions for saving Bauxdorf.

extra software skills beyond those that have to be expected after successfully finishing 6th High School grade in the Austrian educational system.

3.2. Direct training

Participants in experimental settings with metacognitive training received a five-minute introduction labelled "Learning to learn" prior to using the game-based simulation. Based on the work by Sitzmann and Ely (2010), participants were introduced to the nature and benefits of cognitive and metacognitive strategies, received examples, and subsequently had to write down examples of how they could apply these strategies (see Fig. 3). The strategies provided were (1) activation of prior knowledge, (2) planning and evaluation of one's own progress, (3) evaluation of the relevance of content for one's own objectives, (4) detecting and resolving ambiguities, and (5) final evaluation of the learning process. While other approaches to train metacognitive strategies usually take a couple of units (e.g., Carretti et al., 2014; Fritz & Hussy, 2001; Lucangeli et al., 1995), there is also evidence that short-time interventions can contribute to acquisition and practice of metacognitive strategies (Moser et al., 2017).

3.3. Prompting

In experimental settings with prompting at different stages of the game, participants were required to use the proposed strategies and write down on a sheet of paper provided how they would apply these strategies based on the information provided in the game. These strategies were the same as in the training and started with activation of prior knowledge (prompt 1: cognitive), planning and evaluation of one's own progress (prompts 2–4: metacognitive and cognitive), evaluation of the relevance of content for one's own objectives (prompt 5: metacognitive), detecting and resolving ambiguities (prompt 6: metacognitive), reflecting (prompt 7: metacognitive), repetition and

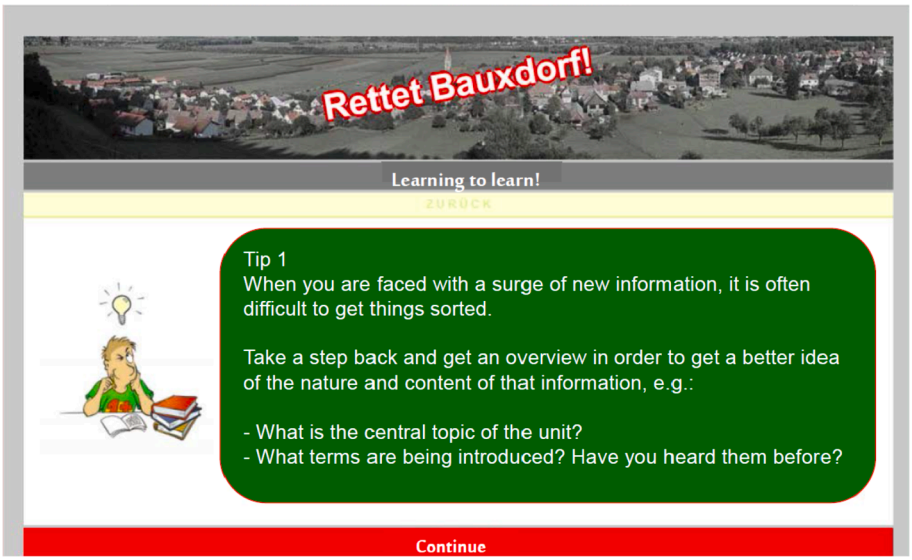


Fig. 3. Sample screen from the direct training on metacognitive learning strategies.

elaboration (prompt 8: cognitive), and final evaluation of the learning process (prompt 9: metacognitive and cognitive; see Fig. 4).

The prompts comprise the activation of the use of metacognitive as well as cognitive strategies. Table 1 shows a list of all nine prompts, each with examples of the different strategies learners had to use and think about.

3.4. Instruments and data analysis

In order to assess participants' knowledge before and after the intervention, a test comprising 12 multiple choice questions (e.g., "What is the average age in Austria?" – 35.2/68.1/28.4/**43.2**) and 5 open questions (e.g. "What would you do with regard to the ageing population in Bauxdorf to meet the requirements of older people and make the village attractive for senior citizens?") was developed. The multiple choice questions were mainly designed to assess basic *factual knowledge* while the open questions were designed to assess *understanding* of the learning material. For assessing cognitive load, the NASA-TLX questionnaire was administered in the post-test (Hart & Staveland, 1988, 6 items, *Cronbach's Alpha* = .70; e.g., "The learning task was difficult"). The repertoire of cognitive and metacognitive learning strategies was measured with the LIST questionnaire (Wild & Schiefele, 1994) using the following subscales: elaboration (e.g., 8 items, "*Cronbach's Alpha* = .68; e.g., "I try to find an example for practical application when faced with a new concept"), critical analysis (8 items, *Cronbach's Alpha* = .77; e.g., "I ask myself whether new information I am presented with is convincing"), and metacognitive strategies (11 items, *Cronbach's Alpha* = .82; e.g., "When faced with new objectives, I try to figure out in advance which parts I have to learn and which not").

All scales and subscales were aggregated for data analyses using mean values of single items. Results from knowledge tests were aggregated using sum scores. For data analysis purposes, a MANCOVA using Training and Prompting as independent variables was computed for analyzing main effects. The results from the knowledge post-test (i.e., the overall score from the multiple choice test and the open questions) and Cognitive Load were used as dependent variable. Covariates included the results from the knowledge pre-test (again the overall score from the multiple choice test results and the open questions) and the three LIST scales (Elaboration, Critical Analysis, Metacognitive Strategies). An additional ANOVA with repeated measurement was computed

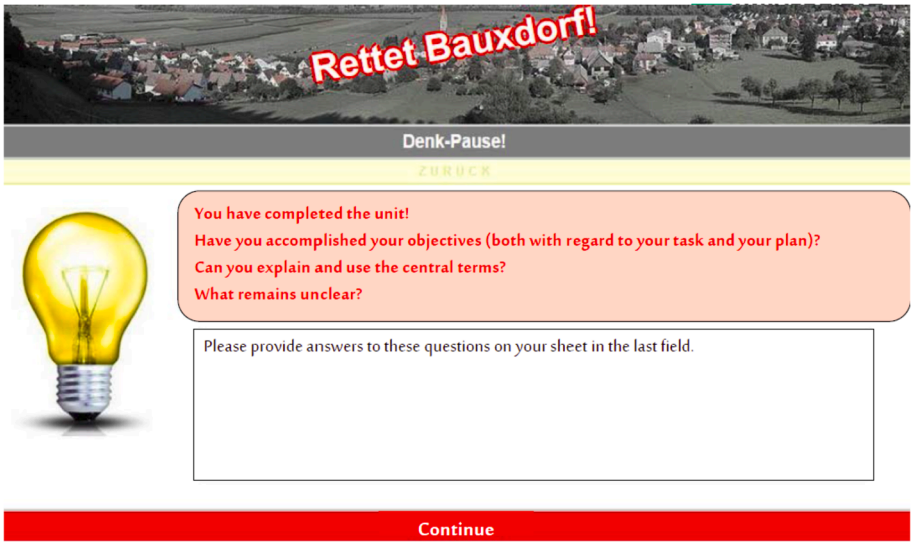


Fig. 4. Metacognitive and cognitive prompting.

Table 1
List of metacognitive and cognitive prompts.

Type of Prompt	Example
Prompt 1: • activationof prior knowledge • cognitive	Before you start learning, think about what you already know about demographic change. • Have you heard about it before? If so, what topics and terms can you remember?
Prompts 2–4: • planning and evaluation of one's own progress • metacognitive and cognitive	Think about what is required from you before you start the game. • What is the stated problem? • How do you plan to solve the problem? • What is your objective? • Can you remember a similar learning situation? If so, think of how you have solved that problem in the past. Please note the central terms of what you have seen. • What is the central theme? • What did you understand and what did you not understand? • Have you ever heard about "push & pull-factors"? • Which other geography topics is this related to?
Prompt 5: • evaluating the relevance of content for one's own objectives • cognitive	Which information is essential to the problem and your objectives, including for your further progress?
Prompt 6: • detecting and resolving ambiguities • metacognitive	You have already made some important decisions. How did you decide on your problem solving strategies?
Prompt 7: • reflecting • metacognitive	Take a short break and think about your progress. • Did you follow your plan? • Have you tried to accomplish your objectives and tasks successfully?
Prompt 8: • repetition and elaboration • cognitive	You have probably learned some new terms. Name the ones you remember and try to provide examples, e.g., migration within the country – a family moves from Salzburg to Linz.
Prompt 9: • evaluation of the learning process • metacognitive and cognitive	You have completed the unit! • Have you accomplished your objectives (both with regard to your task and your plan)? • Can you explain and use the central terms? • What is still unclear?

to analyses differences from pre to post test.

4. Results

With regard to learning outcomes, the descriptive data reveals that participants performed better in the post-test than in the pre-test across all experimental settings (see Table 2). A single ANOVA with repeated measurement reveals that this difference is significant $F(1, 130) = 685.02$, $p < .001$; $\eta_p^2 = 0.84$.

With regard to the first research question about the positive impact of training, the descriptive results reveal that there is hardly any difference between settings that received training prior to exposure to the learning environment and those who did not (see Table 1). This difference is also not statistically significant ($F(1, 123) = 0.002$, $p = .95$; $\eta_p^2 < 0.01$). However, a more detailed look into the different categories of knowledge assessment (multiple choice and open questions) revealed a slight positive effect of training on performance in the multiple choice post-test ($F(1, 122) = 4.10$, $p = .045$; $\eta_p^2 = 0.03$), but not for open questions ($F(1, 123) = 1.38$, $p = .24$; $\eta_p^2 = 0.01$). There was also no effect on cognitive load although participants within the "Training" setting reported a smaller average score here ($F(1, 123) = 0.55$, $p = .46$; $\eta_p^2 < 0.01$).

The second research question addressed the impact of prompting as applied in the experiment. The descriptive results reveal that participants in the prompting group achieved higher overall scores in the knowledge post-test and reported a lower cognitive load. This overall effect is statistically significant ($F(2, 122) = 34.44$, $p < .001$; $\eta_p^2 = 0.36$). A closer look at between-subject effects reveals that prompting had a significant effect on both types of knowledge assessment (multiple choice post-test: $F(1, 122) = 64.57$, $p < .001$; $\eta_p^2 = 0.35$; open questions: $F(1, 122) = 36.91$, $p < .001$; $\eta_p^2 = 0.23$). Again, there was no effect on cognitive load although participants with prompting achieved a lower average score ($F(1, 122) = 0.96$, $p = .33$; $\eta_p^2 < 0.01$).

Table 2
Mean values (and standard deviations in brackets).

	Experimental Condition			
	with Training (n = 70)	without Training (n = 61)	with Prompting (n = 62)	without Prompting (n = 69)
Overall Knowledge pre-test	11.84 (4.50)	11.45 (4.60)	11.72 (4.11)	11.60 (4.91)
Overall Knowledge post-test	22.05 (4.24)	22.05 (3.98)	24.35 (2.79)	20.02 (4.03)
Multiple Choice pre-test	6.26 (2.08)	6.10 (1.92)	6.10 (1.84)	6.26 (2.15)
Multiple Choice post-test	10.67 (1.84)	10.20 (1.80)	11.45 (1.30)	9.55 (1.78)
Open Questions pre-test	5.58 (3.17)	5.35 (3.41)	5.62 (3.25)	5.34 (3.32)
Open Questions post-test	11.41 (3.10)	11.85 (2.70)	12.90 (2.15)	16.47 (3.05)
Metacognitive Strategies	3.56 (.85)	3.54 (.72)	3.56 (.81)	3.54 (.78)
Elaboration	3.19 (.67)	3.28 (.65)	3.26 (.61)	3.20 (.71)
Critical Analysis	2.81 (.82)	2.63 (.67)	2.62 (.78)	2.62 (.73)
Cognitive Load	3.00 (.71)	3.14 (.72)	3.03 (.78)	3.10 (.66)

Finally, an interaction of both independent variables was hypothesized. However, the results of the MANCOVA did not show a significant interaction effect ($F(2, 122) = 0.32$, $p = .32$; $\eta_p^2 = 0.02$).

Finally, the impact of control variables on outcomes was examined. Here, the results show a significant effect of prior knowledge on knowledge post-test ($F(1, 123) = 38.01$, $p < .001$; $\eta_p^2 = 0.24$) and an effect for critical analysis on cognitive load ($F(1, 123) = 4.79$, $p = .03$; $\eta_p^2 = 0.04$), but no other significant effects. Correlation analysis shows a positive correlation between the overall knowledge scores pre- and post-test ($r = 0.45$; $p < .001$), while critical analysis and cognitive load correlate negatively ($r = -0.22$; $p = .01$).

5. Discussion

While recent research has shown that game-based learning with more or less serious games can be effective for learning, other research has shown that such instructional approaches sometimes fail. What is more, research about different scaffoldings approaches within the cognitive as well as metacognitive domain have shown that such support strategies are sometimes effective and sometimes not (Bannert, 2007; Garner & Alexander, 2015; Glogger et al., 2008).

This study analyzed how a serious game can be used for teaching students the basic concepts of demographic change within the scope of geography lessons. The topic itself can be regarded as rather "ill-structured", and is characterized as a topic where multiple perspectives paint different and contradictory pictures. The chosen approach with a serious game for exploring the domain and finding possible solutions to problems proved efficient. The results revealed that the learning environment contributed to knowledge acquisition across all experimental settings from pre-to post-test. However, the underlying research questions addressed issues of supporting learning with simulation-based serious games by fostering cognitive and metacognitive processes (cf.

Wouters & Van Oostendorp, 2013). Two basic supporting mechanisms and their interactive influences on knowledge acquisition and cognitive load were applied and examined. The first approach chosen here was a rather direct approach involving training on how to activate and apply cognitive and metacognitive strategies. The effects of this method proved to be weak and ambiguous: we only found a weak effect on memorizing individual concepts and facts as assessed by a multiple choice test, but no overall effect on open questions aimed at testing understanding. With regard to cognitive load, there was a small but non-significant difference showing that participants who had received training reported a lower cognitive load. These findings indicate that training might support the use of cognitive elaboration processes, but not the application of metacognitive reflective processes. A major reason for this could be the duration of the training, which was a brief rather than a broad introduction into nature, purpose, and application of metacognitive learning strategies. Although there is work that shows that even such short-time trainings can have substantial impact on the use of metacognitive strategies (e.g.,)(Moser et al., 2017), there is also evidence that trainings that have proven effective in this respect indeed involved longer training times and more opportunities to practice (e.g., Carretti et al., 2014; Fritz & Hussy, 2001; Lucangeli et al., 1995; Yuill & Oakhill, 1991). Thus, the training provided in this study may have been too short here in order to reveal sustainable effects and to transfer the strategies into the learning environment. It might be beneficial to extend such training approaches, and to also include the application of metacognitive learning strategies within the domain or the learning environment that should be applied in order to increase the probability of transfer. With the training as applied here, it might also be possible that learners acquired basic metacognitive strategies, but did not see a need for practicing them during the game for themselves, because they were simply not required to do except in the prompting condition. As the prompts required to apply the same strategies as used in the training, the training and the prompting might have been redundant here.

The second approach chosen here was the prompting of cognitive and metacognitive strategies. This approach is referred to as "metacognitive prompting" because the activation and application of cognitive strategies is itself a metacognitive process. With regard to the effects of this prompting, the results reveal a strong impact on knowledge acquisition. It must be noted here that the use of the prompts provided was mandatory, so that every participant had to apply them within predetermined stages of the learning game. The findings of other studies, where the use of such prompts was not mandatory, show that they are only effective when used properly (e.g., Garner & Alexander, 2015; Moser et al., 2017). In these studies, prompting was effective for learning but had no influence on cognitive load (neither positive nor negative). While it could be assumed that the use of such prompts takes learners "out" of the learning context (cf. Clark et al., 2014), the data does not suggest such an effect. Thus, the provision and use of this scaffolding technique worked well here and did not interrupt the gameplay, but rather became part of the learning environment.

Finally, an interaction effect between both supporting approaches was hypothesized. Nevertheless, the results did not reveal such an effect. A major reason for this may be that the training did not work as intended, leading to a shortage of variance that could have interacted with the prompting. In addition, the results with regard to the covariates also suggest that the availability of metacognitive strategies does not automatically lead to their application (cf. Bannert, 2007). There was no statistically significant impact of the repertoire of metacognitive strategies as assessed in the pre-test. The only covariates that had an impact on post-test performance were prior knowledge and the use of critical analysis strategies. The moderate positive correlation between pre-test and post-test performance in the knowledge test is not surprising but is a pattern found in many studies using a similar design (e.g., Hattie, 2013; Moser et al., 2017). The weak negative correlation between critical analysis and cognitive load is also not surprising, because learners who are used to questioning themselves invest greater mental effort and thus need more capacity for cognitive activities.

Taken together, research on serious games has shown that in most cased learners benefit form additional instructional support (Wouters & Van Oostendorp, 2013). Unclear was how metacognitive processes during learning with serious games can be fostered. There is evidence that metacognitive prompting (e.g., Bannert et al., 2009) and training (Carretti et al., 2014) can support students during self-regulated learning, but how these instructional approaches can support learning from and with serious games within the domain of conceptual change remained unclear. While research has shown that self-reflection as one instance of metacognitive monitoring is beneficial for learning with serious games (Riemer & Schrader, 2016), the supportive strategies as applied here are new within this field of research. Outcomes show that the training as applied here did not really contribute to an increase in learning performance, but the metacognitive prompting is an effective instructional support within the kind of serious games that has been developed for this study.

5.1. Limitations of the study

The limitation of this research primarily addresses ecological validity of the chosen approach here. It has to be noted that the study has been conducted under controlled conditions. Thus, students were actively encouraged to use the prompts as provided by the program. Prior research (e.g., Moser et al., 2017) has shown that metacognitive prompting is only beneficial for learning when the prompts are used adequately. With regard to metacognitive training, it is also obvious that the short-time intervention as used here does not really result in immediate effects. Here a long-term intervention might be more promising (e.g., Carretti et al., 2014). In addition, the domain or content might play a crucial role whether metacognitive support is beneficial or not. The ill-structured domain used within this study seems to represent a field where multiple sources and perspectives have to be evaluated. Thus, the scaffolding as applied here seems to be a beneficial supportive strategy. Within other, more simple and linear domains this kind of support might not be helpful or even increase learners' cognitive load by unnecessarily increasing task complexity. Finally, no process data (e.g., eye-tracking, log-files, etc.) was obtained in this study. Thus, no information about students' cognitive and metacognitive processes in the background is available here in order to draw conclusions beyond the results as provided by the aggregated data here.

6. Conclusions

Overall, the outcomes of this study suggest that learning with serious games benefits from additional instructional support by means of scaffolding. The provision of metacognitive prompting was effective in this respect and enhanced knowledge acquisition without simultaneously increasing cognitive load. Sharing metacognitive learning strategies with students might also support learning but must be extended in terms of both time and practice. When using serious games in classroom practice instructors have to concisely evaluate how available games meet the objectives of the underlying curricula and how these applications can support students in reaching these objectives or standards. Outcomes of this study suggest that additional instructional support like metacognitive scaffolding can facilitate this process.

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The authors declare that they have no conflict of interest.

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